

Development Roadmap

Phase 1 – Research & Planning

Drone architecture study

Component selection

BOM creation

Repo setup

Wiring diagrams

Phase 2 – Custom Controller

Build transmitter PCB/prototype

Read joysticks

Send RF signals

Power testing

Phase 3 – Receiver System

Receive commands

Decode packets

Safety failsafe system

Phase 4 – Flight Controller

Read IMU data

Calibrate sensors

PID stabilization

Motor mixing

Phase 5 – Airframe Build

Frame assembly

Wiring integration

ESC + motor installation

Phase 6 – Testing

Bench tests

Motor tests

Hover tuning

Flight trials

Phase 7 – Documentation Release

Final code upload

Build guide

Open-source release

Why This Project Matters

Many people use ready-made drone kits without understanding how they work.

This project breaks the system down and rebuilds it from the ground up, helping students and makers learn:

How drones balance

How wireless controllers work

How sensors guide movement

How embedded firmware controls hardware

Current Status

● Planning & Initial Design Stage & building

Future Improvements

GPS waypoint mode

Camera module

FPV support

Mobile telemetry app

Auto landing

Obstacle sensing
